SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN

COURSE CODE: CSA1104

1. Use Case Diagram Scenario

A Smart Pharmacy Management System enables customers to search medicines, upload

prescriptions, place medicine orders, and make online payments. Pharmacists verify

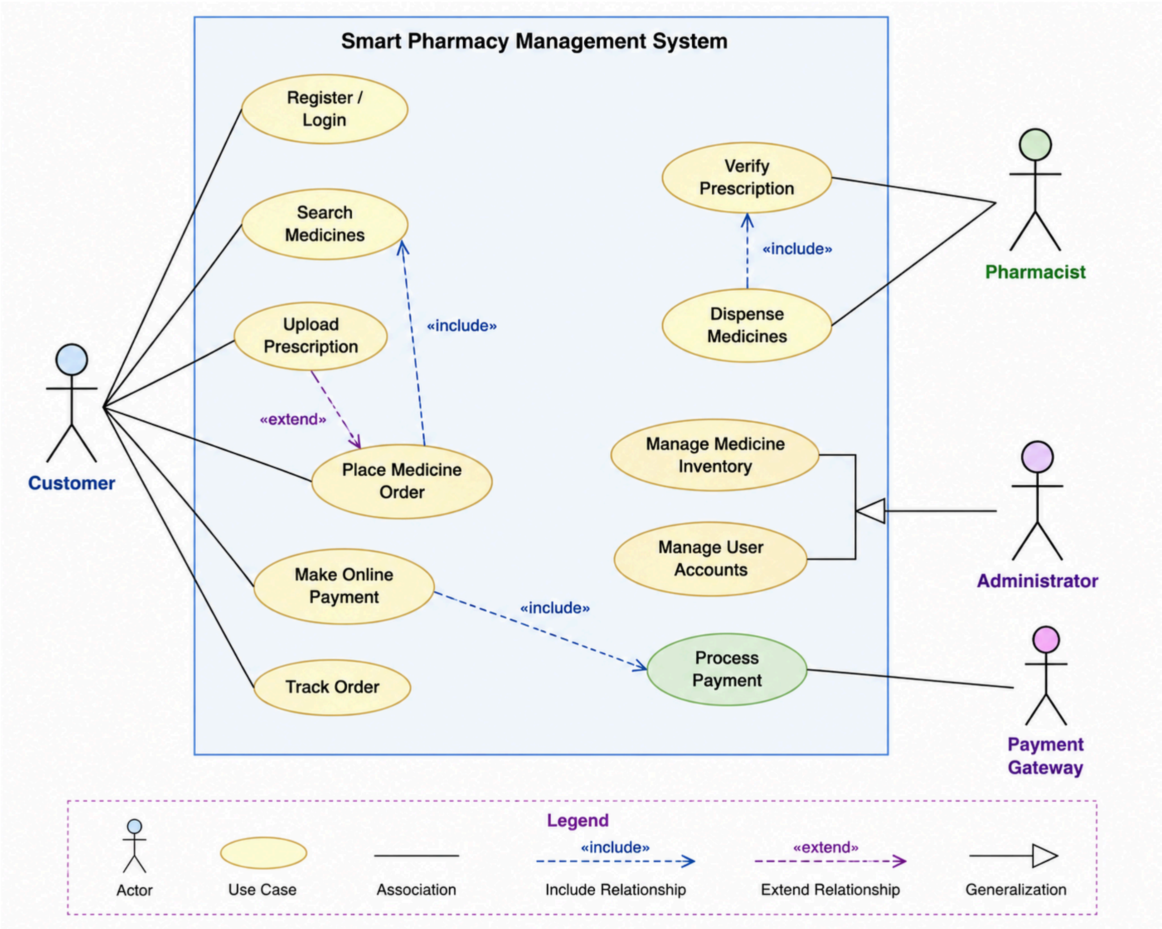
prescriptions and dispense medicines, while administrators manage medicine inventory and

user accounts. The interactions among system users are not clearly defined.

Based on this scenario, analyze the system requirements, identify all actors and use cases, and

design a Use Case Diagram showing appropriate relationships (include, extend, and

generalization) to ensure complete sy



stem functionality.

2. Class Diagram Scenario

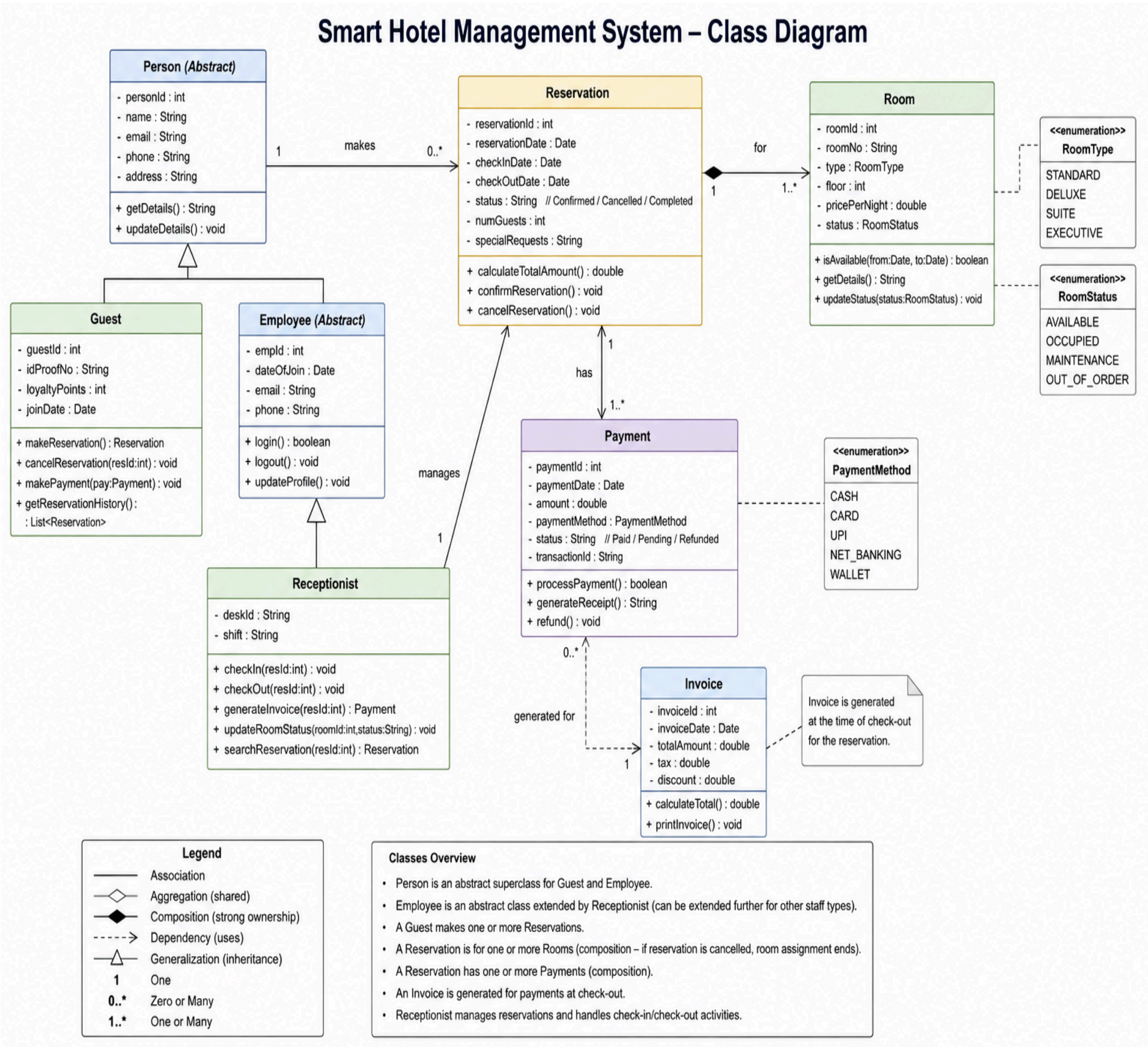
A Smart Hotel Management System consists of entities such as Guest, Room, Reservation,

Payment, and Receptionist. The existing design does not clearly define class attributes,

operations, or relationships among these entities.

Based on this scenario, identify the required classes, define suitable attributes and methods,

establish relationships (association, aggregation, composition, and inheritance), specify multiplicity, and construct a detailed Class Diagram.



3. Sequence Diagram Scenario

In an Online Banking System, a customer logs in, transfers funds, verifies the transaction

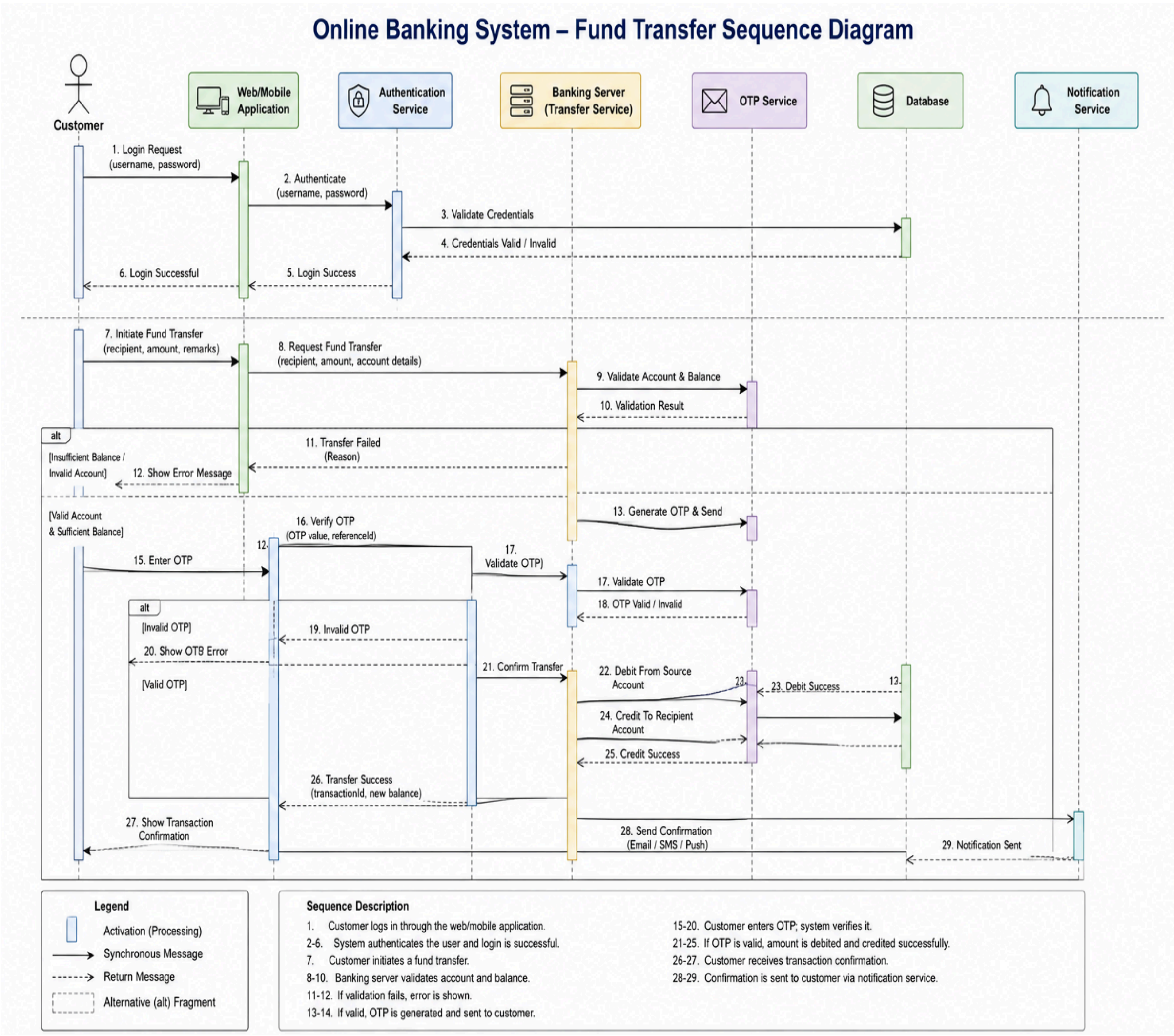
using OTP, and receives a transaction confirmation. The communication among the customer,

authentication service, banking server, and database is incomplete.

Based on this scenario, analyze the sequence of operations, identify participating objects and

message flows, and design a Sequence Diagram representing the complete interaction

process.

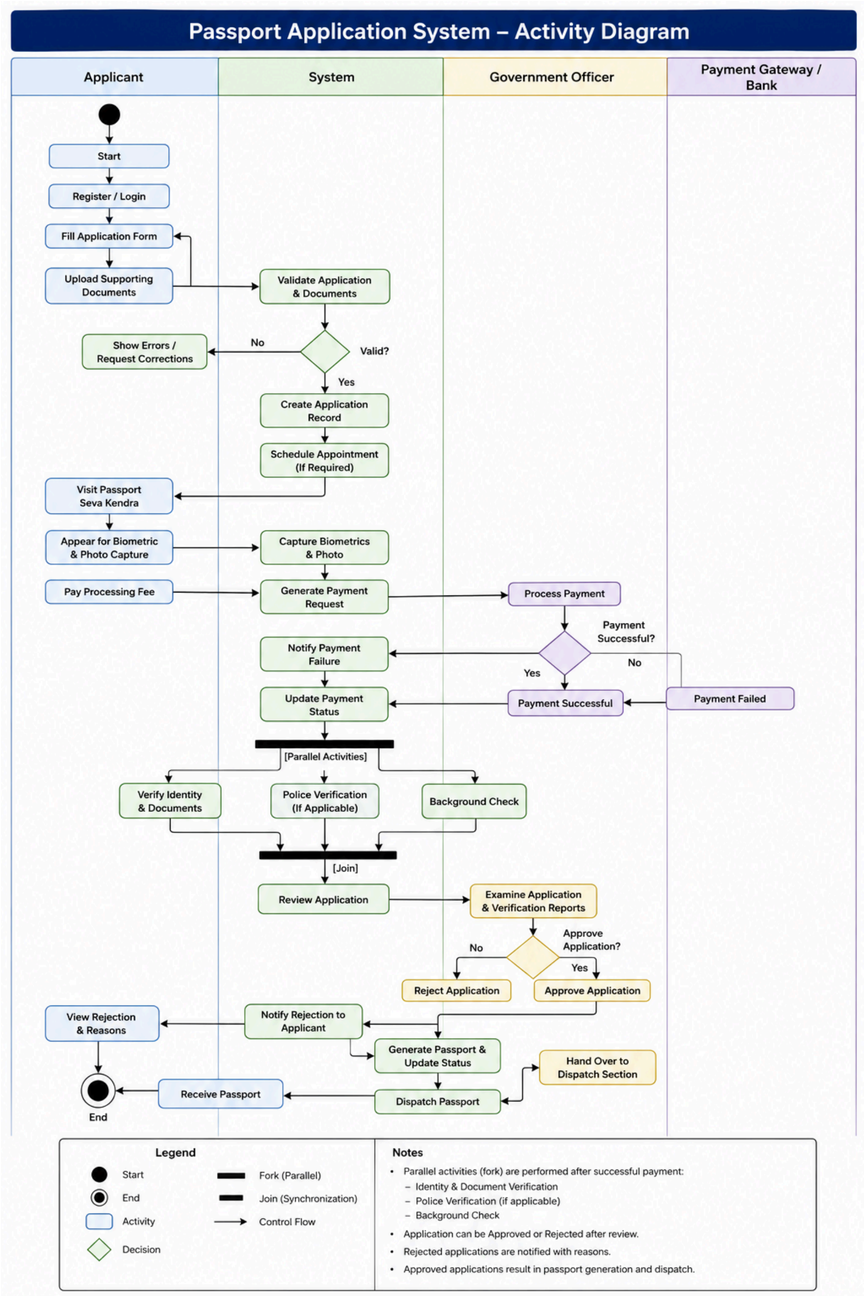


4. Activity Diagram Scenario

A Passport Application System allows applicants to submit applications, upload supporting

documents, complete identity verification, pay processing fees, and receive passports. The workflow does not clearly represent approval, rejection, or concurrent activities. Based on this scenario, analyze the workflow, identify decision points and parallel activities, and design an optimized Activity Diagram using decision nodes, forks, and

joins.



5. State Diagram Scenario

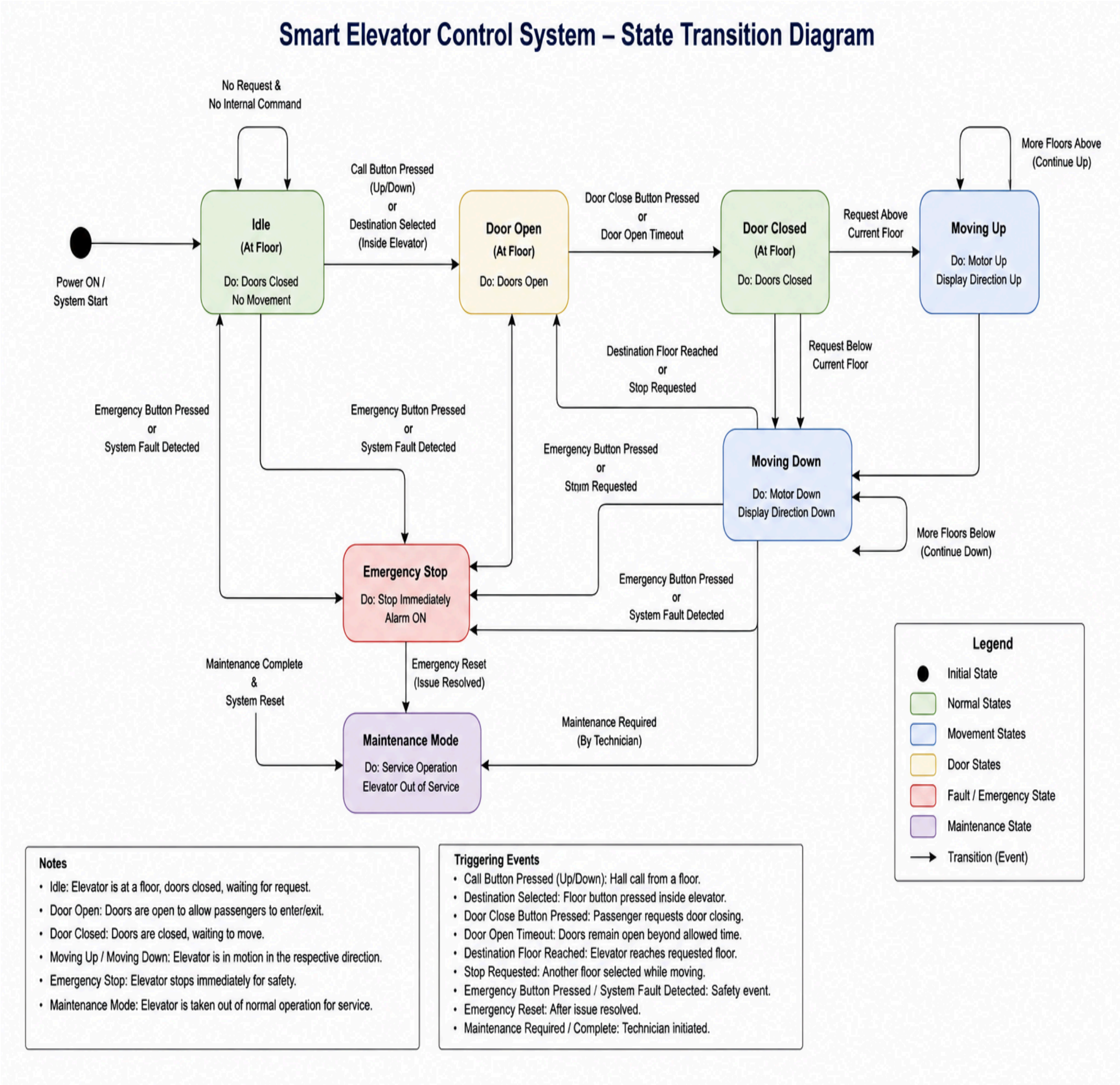
A Smart Elevator Control System operates through states such as Idle, Door Open, Door

Closed, Moving Up, Moving Down, Emergency Stop, and Maintenance Mode. The

transitions between these states are not clearly specified.

Based on this scenario, identify all possible states, triggering events, and transitions, and

construct a State Transition Diagram that accurately represents the elevator& behaviour.



6. Deployment Diagram Scenario

A Smart Home Automation System uses smart sensors, IoT controllers, a home gateway,

cloud servers, and a mobile application to monitor and control lighting, security, and

appliances remotely. The deployment architecture does not clearly illustrate hardware nodes, deployed software, or communication channels.

Based on this scenario, analyze the deployment environment, identify all hardware nodes and

software artifacts, and design a Deployment Diagram showing the physical architecture and communication links.